

A Graph-based Framework for Coverage Analysis in Autonomous Driving

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Motivation

Coverage analysis is crucial to ensure safety and reliability of autonomous driving systems

Coverage argument: have the observed scenes exhausted the space of traffic scenes that can occur?

Existing arguments are typically collected **per coverage factor**, or up to 2–3 factor interactions

This work: take the **complete traffic scene** as the unit of coverage
each scene $\rightarrow$ one graph

nodes carry actor parameters, edges carry relation parameters

Arbitrarily complex scenes—varying numbers of actors, higher-order interactions—captured in one object, not decomposed into isolated factors

Scope: completeness of observed scenes, *not* safety of the system

Existing Coverage and Analysis Approaches

Scenario-based testing is the dominant paradigm (PEGASUS: functional / logical / concrete scenario levels)

Approval trap: statistically demonstrating superior safety needs billions of test km
→ motivates simulation and systematic coverage methods

Standardized coverage criteria: ISO 21448, UL 4600; domain-specific coverage buckets (Foretellix)

Recent trend: represent traffic scenes as **graphs** for scalable, label-free analysis (semantic scene-graph embeddings, contrastive learning, heterogeneous scenario graphs)

Gap: these graph representations are not designed specifically for coverage analysis
→ this work

Argoverse 2.0: large-scale real-world dataset

250,000 scenarios, 11 s each; tracked trajectories (vehicles, pedestrians, cyclists) + semantic map

subset of 17,000 training scenes used

CARLA 0.9.15: open-source simulator

six maps, 20–60 vehicles per run, mixed traffic

2,050 scenes of 11 s each

Defining a Traffic Scene Graph

Map graph G_{map} : directed multigraph, nodes = lane segments (geometry + semantics)

edges: **following**, **neighbor**, **opposite**

Actor graph G_{actor} : dynamic relationships between actors at one timestep

nodes: primary/occupied lane IDs, position s , 3D position, speed, actor type, lane-change indicator

edges: actor-actor relation type + path length along the lane network

Three relation types, ordered by semantic hierarchy: **Following/Leading**, **Neighbor**, **Opposite**

Lane-change indicator ($\Delta t = 1\text{ s}$) already captures cut-in/cut-out within a single-timestep graph

Hierarchical Graph Construction Algorithm

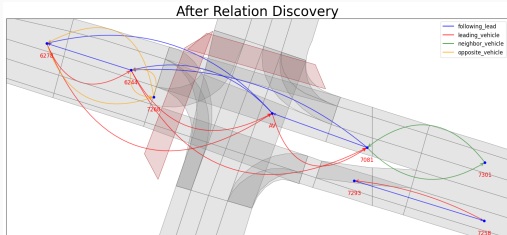
Phase 1 – Relation Discovery: for each actor pair, find a connecting path in the map graph; path structure determines relation type; filtered by distance/path-length thresholds

Phase 2 – Hierarchical Construction: add edges leading/following → neighbor → opposite (shortest path first); skip an edge if a path already exists (BFS)

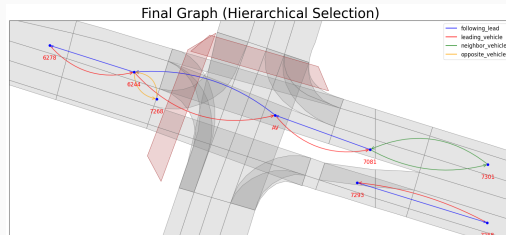
Relation	Max. distance fwd/bwd [m]	Max. path length [edges]
Leading/following	100	3
Neighbor	50 / 50	2
Opposite	100 / 10	2

Redundancy pruning reduces edges while preserving connectivity, keeping isomorphism checks tractable in dense traffic.

Hierarchical Construction: Example



After relation discovery



After hierarchical selection

The discovery graph (left) contains all relations within distance limits. Hierarchical selection (right) removes redundant edges representable through existing paths.

Coverage Method 1: Subgraph Isomorphism

Traffic scene archetypes (lead-follow, cut-in, cut-out, opposite traffic, platoons, ...) expressed as small graphs

Check whether a subgraph of scene graph G is isomorphic to archetype A (VF2 algorithm)

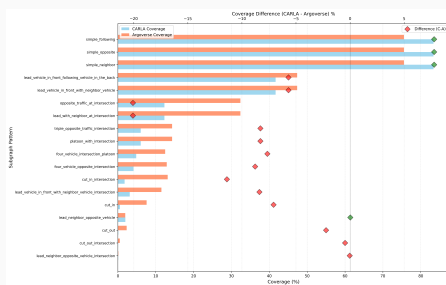
Strategy: (1) define archetype set S ; (2) fix node/edge attributes considered; (3) match each scene graph against $S \rightarrow$ coverage table C

18 archetypes defined: 2-actor (3), 3-actor (9), 4-actor (4), 5-actor (2)

Archetypes are user-defined, illustrative—not a complete or safety-certified taxonomy

Real safety case: archetypes derived top-down (HARA / UL 4600), completeness of the set is critical

Results: Scenario Archetype Coverage Gaps



CARLA higher coverage only for the 3 simple 2-actor scenarios

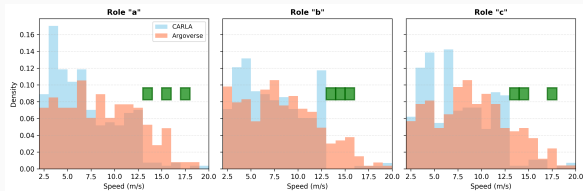
All multi-actor scenarios: Argoverse shows higher occurrence

cut_out_intersection: <1% CARLA vs. ~8% Argoverse

lead_neighbor_opposite_vehicle_intersection: ~6% Argoverse, virtually absent in CARLA

cut_in/cut_out: underrepresented in CARLA regardless of intersections

Results: Parameter Distribution Gaps



Role-specific speed distributions,
lead_vehicle_in_front_with_
neighbor_vehicle scenario

CARLA concentrates around 3–5 m/s
Argoverse extends to 15–20 m/s and
beyond

Coverage gap in the 13–17 m/s range
for all three roles

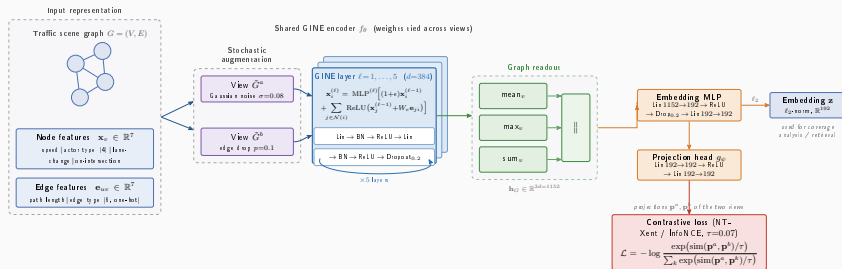
Gaps arise at the parameter level even
within structurally matching scenarios

Coverage Method 2: Graph Embeddings (GINE)

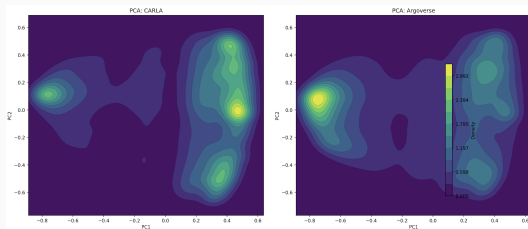
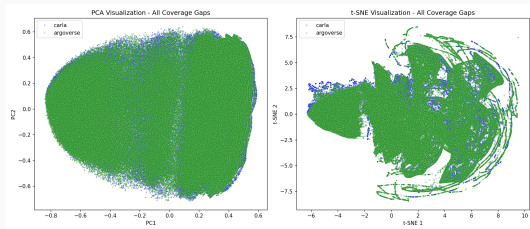
Real traffic scenes are far more complex than predefined archetypes → complementary top-down approach

Graph Isomorphism Network with Edge features (GINE): natively encodes edge attributes (relation type + distance)

Self-supervised contrastive training (NT-Xent/InfoNCE, $\tau = 0.07$) on CARLA + Argoverse jointly



Results: Embedding Space Analysis

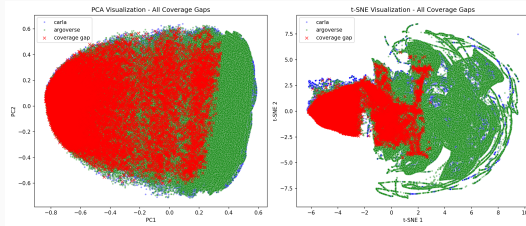


Plausibility check: nearest neighbours in embedding space share actor configuration and road geometry

First two principal components explain 40% of the total variance

CARLA peaks at higher PC1 values; Argoverse peaks at negative PC1 values → datasets occupy different regions of the embedding space

Results: Coverage Gaps in Embedding Space



Argoverse graphs with under-represented archetype labels projected into embedding space and compared against CARLA density

23,148 Argoverse graphs identified as coverage gaps

Dominated by `cut_in` (~94% of Argoverse cut-in instances), `cut_out`, `cut_out_intersection`, `lead_neighbor_opposite_vehicle_intersection`

Gap graphs cluster where CARLA density is negligible → agreement between embedding and exact isomorphism labels

Used as a *coverage-screening* layer, not a standalone safety oracle

Summary and Future Work

Graph-based framework: hierarchical scene graphs combining map topology with actor relationships

Two complementary coverage methods:

- subgraph isomorphism against archetypes → interpretable, pattern-based coverage

- GINE embeddings via contrastive learning → similarity-based coverage, clustering, anomaly detection

Validated on Argoverse 2.0 and CARLA: reveals structural, parametric, and co-occurrence coverage gaps in simulation

Scales across scenarios without scenario-specific handling; naturally accommodates varying numbers of actors

Future work: multi-timestep temporal graphs; automatically extracting new archetype candidates from dense, unmatched embedding regions

Code: https://github.com/tmuehlen80/graph_coverage